

STUDENT PROJECT

★ HOMEWORK

AI-Powered Quiz Generator Application

Build an intelligent system that transforms any PowerPoint into an interactive, scored multiple-choice quiz — powered by AI.

Educators and students struggle to quickly generate meaningful practice quizzes from existing study material. This project addresses that gap by building an AI application that takes a PowerPoint file as input and automatically generates a fully interactive MCQ quiz.

1

Manual Effort

Creating quiz questions from slides is tedious and time-consuming for educators.

2

No Feedback Loop

Students answer questions but receive no explanation of why their answers are incorrect.

3

One-size-fits-all

Existing tools rarely support adaptive difficulty levels tailored to learner needs.

OBJECTIVE

- Accept a .pptx file as AI input source
- Let users choose number of questions (5–30)
- Support three difficulty levels: Simple, Medium, Complex
- Generate well-formed MCQs using AI (4 options each)
- Deliver an interactive quiz interface
- Score answers and explain why wrong options are incorrect

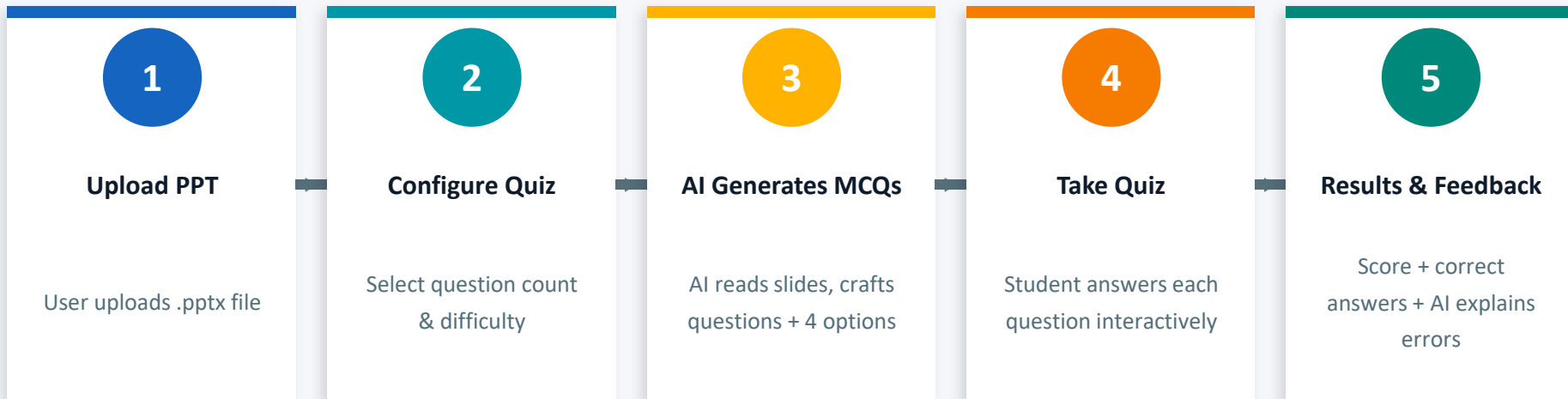
SCOPE

IN SCOPE

- ✓ PPT/PPTX file upload
- ✓ AI MCQ generation
- ✓ Difficulty level filtering
- ✓ Timed interactive quiz UI
- ✓ Score + detailed AI feedback

OUT OF SCOPE

- ✗ Video/Audio files
- ✗ Essay/open-ended questions
- ✗ User authentication system
- ✗ Multi-language support



Key Technical Considerations

-  Parse PowerPoint content using python-pptx or similar libraries to extract slide text
-  Leverage an LLM (e.g., Claude, GPT) to generate MCQs with varied difficulty from extracted text
-  Build an intuitive React or HTML/CSS front-end with real-time quiz interaction and feedback display

FR-01

PPT File Input

- Accept .pptx uploads via file picker
- Extract all text content from each slide
- Display slide count and content preview

FR-02

Quiz Configuration

- Input: number of questions (min 5, max 30)
- Dropdown: Simple / Medium / Complex difficulty
- 'Generate Quiz' button to trigger AI processing

FR-03

MCQ Generation

- AI generates questions relevant to slide content
- Each question has exactly 4 options (A–D)
- One correct answer; distractors are plausible

FR-04

Quiz Interface

- Display one question at a time (or paginated)
- Allow single-answer selection per question
- Submit/Next button to progress through quiz

FR-05

Scoring & Feedback

- Show final score (X / Total) upon completion
- Highlight correct answer vs. student selection
- AI-generated explanation for each wrong answer

FR-06

UX Requirements

- Clean, mobile-friendly responsive UI
- Loading indicator during AI generation
- Option to retake quiz or upload new PPT

★ HOMEWORK

UI Mockups

How Your App Should Look

Reference wireframes for every screen of the Quiz Generator. Match the layout, controls, states and feedback shown on the next four slides.

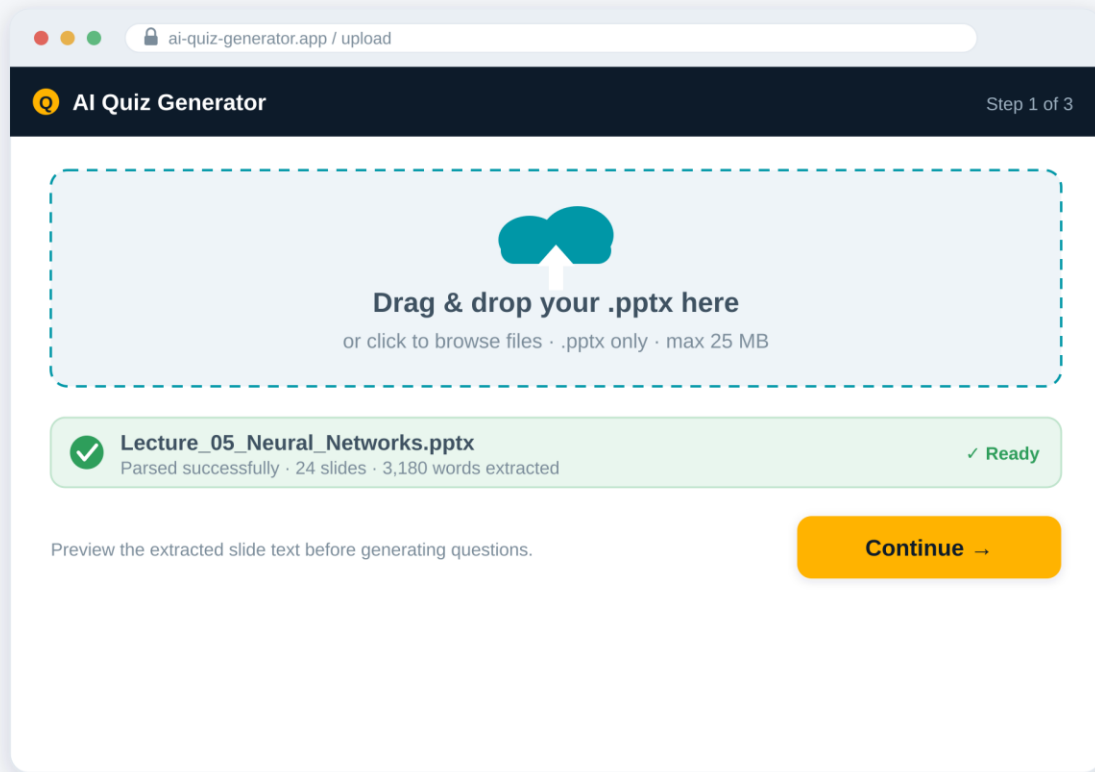
1 Upload & Parse

2 Configure Quiz

3 Take Quiz

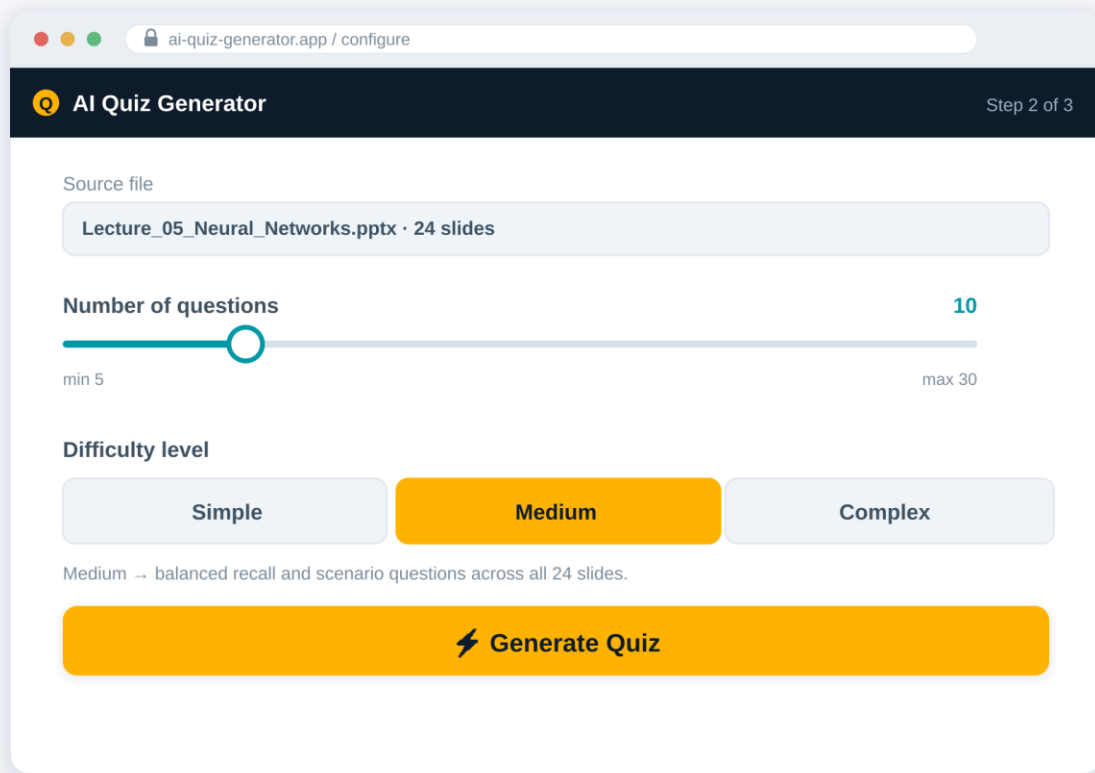
4 Results & Feedback

Build the screens to match these references — they define your grading rubric for UI / UX.



WHAT TO BUILD

- 1 File picker**
.pptx upload via drag-drop or a Browse button.
- 2 Validation**
Reject non-pptx files; show a clear error state.
- 3 Slide count + preview**
Confirm parse worked: show slide count and a content preview.
- 4 Enable next step**
Keep Continue disabled until a file is parsed successfully.



The image shows a web browser window with the address bar displaying 'ai-quiz-generator.app / configure'. The page has a dark header with the 'AI Quiz Generator' logo and 'Step 2 of 3'. The main content area is white and contains the following elements:

- Source file:** A light blue box containing the text 'Lecture_05_Neural_Networks.pptx · 24 slides'.
- Number of questions:** A slider control with a teal track. The value '10' is displayed in teal on the right. The range is from 'min 5' to 'max 30'.
- Difficulty level:** Three buttons: 'Simple' (light blue), 'Medium' (orange), and 'Complex' (light blue). The 'Medium' button is selected.
- Feedback text:** Below the buttons, it says 'Medium → balanced recall and scenario questions across all 24 slides.'
- Generate Quiz button:** A large orange button with a lightning bolt icon and the text 'Generate Quiz'.

WHAT TO BUILD

- 1 Question count**
Numeric input or slider, clamped between 5 and 30.
- 2 Difficulty control**
Simple / Medium / Complex — drives the AI prompt.
- 3 Generate action**
One button sends slide text + settings to the LLM.
- 4 Loading state**
Show a spinner / progress while the AI builds the quiz.

ai-quiz-generator.app / quiz

AI Quiz Generator Question 3 of 10

02:14

QUESTION 3

Which technique most directly reduces overfitting by randomly disabling neurons during training?

☐ A L2 weight regularization

☒ B Dropout ✓

☐ C Increasing the learning rate

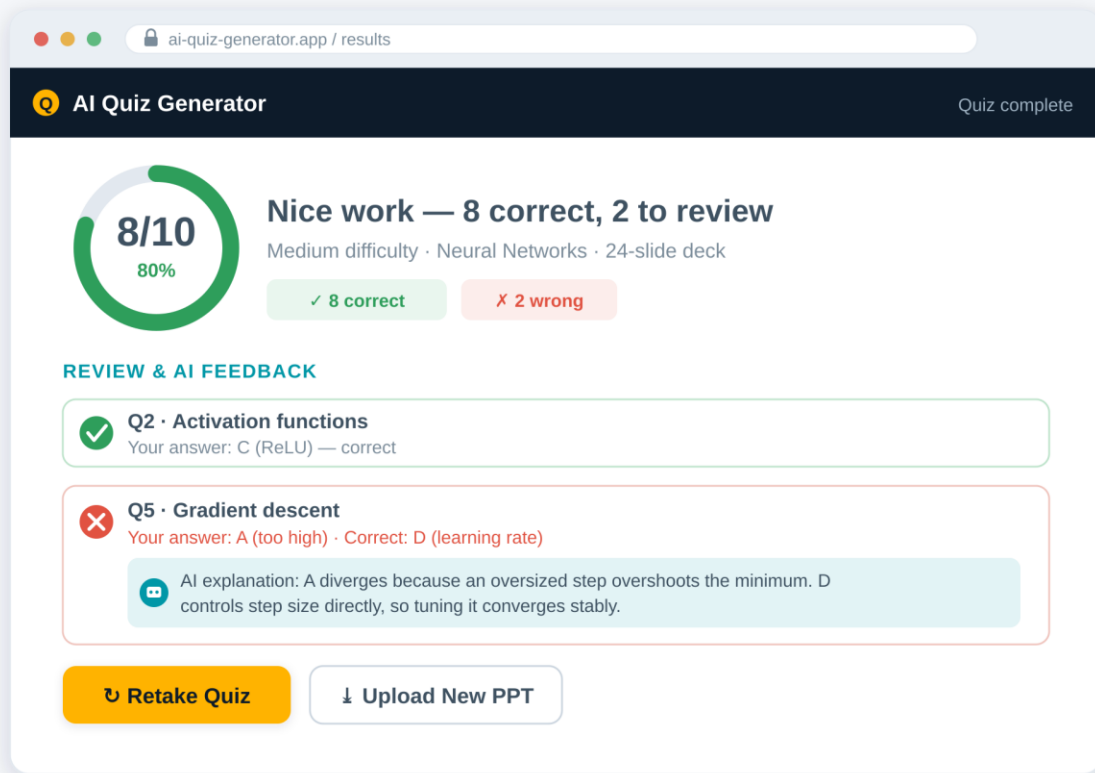
☐ D Adding more layers

← Previous

Next →

WHAT TO BUILD

- 1 One question at a time**
Show the current question with its four options A–D.
- 2 Single selection**
Only one option selectable; highlight the chosen one.
- 3 Progress + timer**
Show position (3 of 10) and an optional countdown.
- 4 Navigation**
Previous / Next, and Submit on the final question.



WHAT TO BUILD

- 1 Final score**
Show score as X / Total and a percentage.
- 2 Answer comparison**
For each question, show the user's pick vs the correct answer.
- 3 AI explanations**
Explain WHY each wrong answer is wrong — the core feature.
- 4 Restart options**
Let users retake the quiz or upload a new deck.

Evaluation Area	Weight	Success Criterion
PPT Parsing & Input	15%	Accurate text extraction from all slides
AI MCQ Generation	25%	Relevant, well-formed questions at correct difficulty
Quiz UI / UX	20%	Responsive, intuitive, and error-free interface
Scoring Accuracy	15%	Correct score computation and answer display
AI Feedback Quality	15%	Clear, meaningful explanations for wrong answers
Code Quality & Docs	10%	Clean code, README, inline comments



Deliverables:

Source code (GitHub) • Working demo / deployed URL • Project report (PDF) • 5-min video demo

Every AI-generated quiz should satisfy each of these attributes so questions challenge real understanding.

1 Make all options plausible

Each option should carry just enough correctness that it can't be dismissed outright — yet only one is truly defensible given every constraint in the stem.

2 Balance length & specificity

The correct answer shouldn't always be the longest. Distractors can be detailed and still miss a crucial regulatory, ethical, or operational fit.

3 Add “best among goods” ambiguity

More than one action may be reasonable, but the question forces the most defensible choice under the given principles — how real exams trip up even experts.

7 Verify every answer key

Ensure the answer marked correct is unambiguously correct — with no defensible second interpretation left open.

4 Use subtle distinctions

Telling the options apart should require genuine subject-matter expertise, not surface-level reading of the question.

5 Cover the whole syllabus

Distribute questions evenly across every topic in the source deck, rather than clustering them around a few slides.

6 Favor scenario-based items

Prefer qualitative, scenario-driven questions that test judgement over simple factual recall wherever possible.



The goal

Reward genuine reasoning, not memorization — a well-prepared student should still have to think.



Ready to Build?

Transform any PowerPoint into an AI-powered quiz experience.

 Input: .pptx

 Process: AI/LLM

 Output: Interactive Quiz

Good luck! Build something amazing. 💡